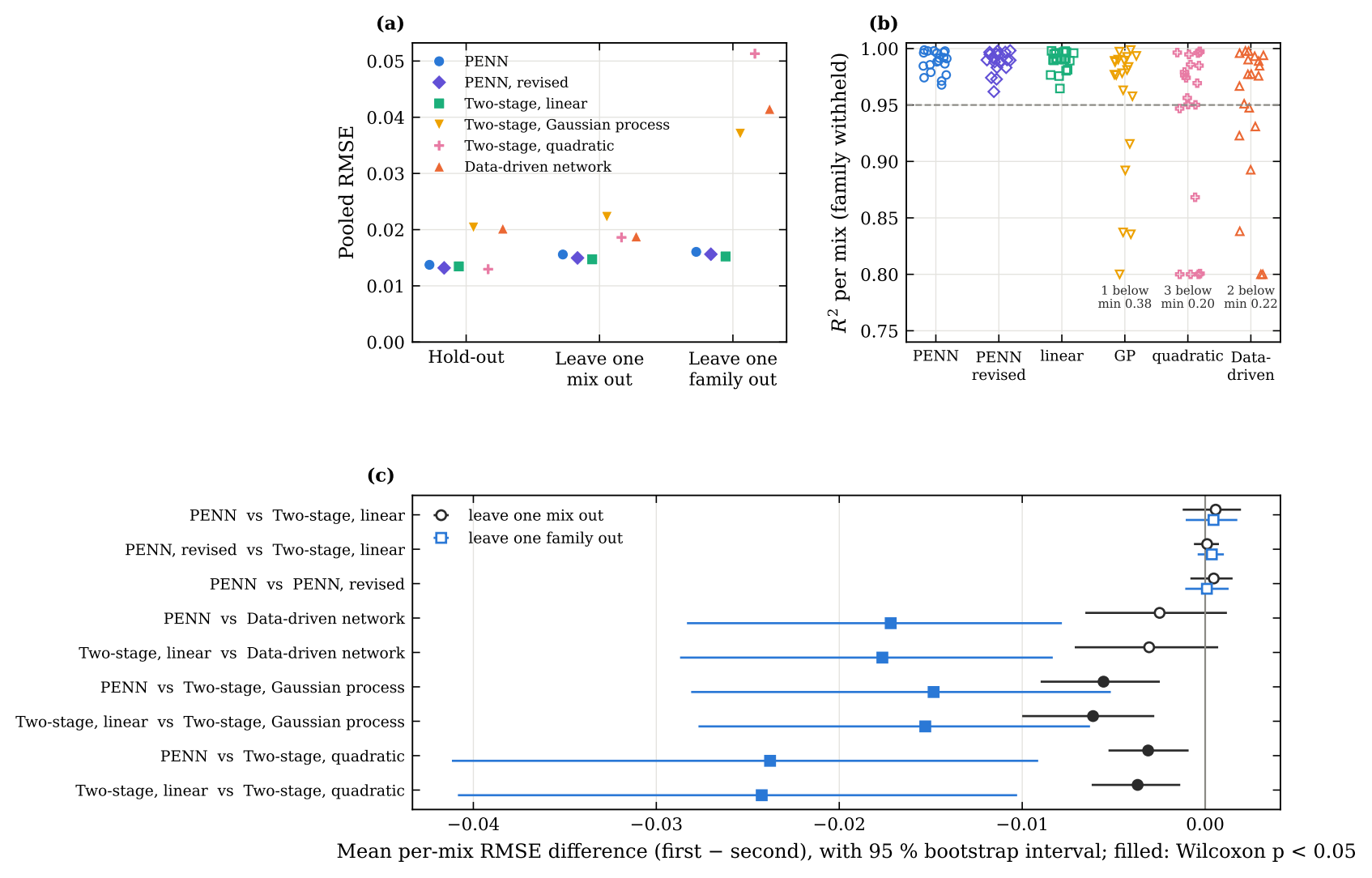
(a)
Pooled RMSE
0.00
0.01
0.02
0.03
0.04
0.05
PENN
PENN, revised
Two-stage, linear
Two-stage, Gaussian process
Two-stage, quadratic
Data-driven network
Hold-out
Leave one mix out
Leave one family out
(b)
R^2 per mix (family withheld)
0.75
0.80
0.85
0.90
0.95
1.00
1 below min 0.38
3 below min 0.20
2 below min 0.22
PENN
PENN revised
linear
GP
quadratic
Data-driven
(c)
PENN vs Two-stage, linear
PENN, revised vs Two-stage, linear
PENN vs PENN, revised
PENN vs Data-driven network
Two-stage, linear vs Data-driven network
PENN vs Two-stage, Gaussian process
Two-stage, linear vs Two-stage, Gaussian process
PENN vs Two-stage, quadratic
Two-stage, linear vs Two-stage, quadratic
leave one mix out
leave one family out
−0.04
−0.03
−0.02
−0.01
0.00
Mean per-mix RMSE difference (first − second), with 95 % bootstrap interval; filled: Wilcoxon p < 0.05